

Q1 Dataset : Hospital Patient segmentation.

Patient ID	Age	Monthly Medical	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	25

SSE values

K	SSE
1	600
2	350
3	220
4	200
5	190

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret following elbow point from the SSE values provided and determine optimal no. of clusters. Discuss affects Patient no. of segmentation and healthcare decision-making.

SSE means $\rightarrow$ sum of Squared Errors.

we are using elbow method.

K	SSE	Reduction from previous K .
1	600	—
2	350	250
3	220	130
4	200	20
5	190	10

When $K=2,3$ the SSE value is reduced much, when $K=4,5$ then decreases very small. This variation happens after the value of $K=3$. Therefore elbow method suggest $K=3$.

Based on health-risk score three clusters are formed;

- low risk with low expense
- Moderate " Moderate "
- High " High "

Impact on decision-making:-

for low risk group you need not be spend more. but for moderate their must be checkups and tips and high risk must have intensive monitoring

The elements within this have small distance, hence this will be one cluster.

Using same notation generate distance matrix:-

Step 4: 0	01	02	03	04	05
00					
01	0	1.697	0.063	1.616	1.663
02		0	1.732	1.423	0.063
03			0	1.647	1.697
04				0	1.412
05					0

Steps: HAC

We use single linkage, where:-

$$D(A, B) = \min d(x, y)$$

between members of two clusters.

Merge 1

Smallest distance:-

$$01 - 03 \Rightarrow 0.0625$$

By seeing smallest distance

$$C_1 = 01 + 03$$

$$C_2 = 02 + 05$$

$$C_3 = 04$$

$$C_1 = \{01, 03\} = 0.0625$$

$$C_2 = \{02, 05\} = 0.0625$$

$$C_3 = \{04\} = 0.0625$$

5Q) A retailer applies Hierarchical Ecommerce order analysis orders using approximately encoded product category information and standardized orders value. A dendrogram is generated. Interpret how clusters are progressively formed and determine the appropriate no. of clusters by selecting a suitable cut-off level on dendrogram. Explain how selected clusters can help

order ID	product category	order value
01	electronics	55000
02	clothing	25000
03	electronics	52000
04	furniture	27000
05	clothing	30000

HAC Algorithm :-

→ calculates distance b/w clusters.

→ Merge two closest clusters.

→ Recalculates the distance.

→ Continue until becomes into one clusters

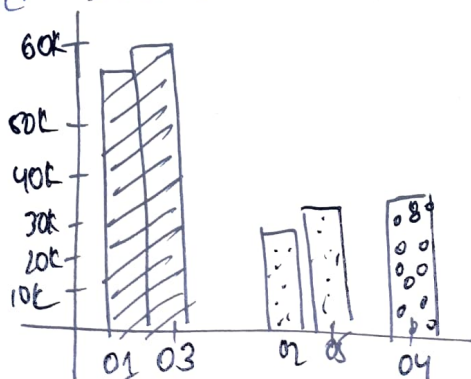
01 and 03 forms one clusters - electronics.

01 - 55000

03 - 52000

02 and 05 - clothing

04 and 3rd cluster - furniture.



But the optional clusters cannot be determined by the given data.

Step 1 Encode the category:

Order	electronics	clothing	furniture
O1	1	0	0
O2	0	1	0
O3	1	0	0
O4	0	0	1
O5	0	1	0

Step 2: Normalize order value.

$$x^1 = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

$$x_{\min} = 25,000$$

$$x_{\max} = 57,000$$

$$O1 = \frac{55000 - 25000}{57000 - 25000} = \frac{30000}{32000} = 0.9375$$

$$O2 = \frac{25000 - 25000}{57000 - 25000} = 0$$

$$O3 = \frac{57000 - 25000}{57000 - 25000} = \frac{32000}{32000} = 1$$

$$O4 = \frac{30000 - 25000}{57000 - 25000} = \frac{5000}{32000} = 0.15625$$

$$O5 = \frac{27000 - 25000}{57000 - 25000} = \frac{2000}{32000} = 0.0625$$

Step 3 calculate Euclidean Distances;

O1 and O3 :-

$$D = \sqrt{(1-1)^2 + 0^2 + 1^2 + 0^2 + (0.9375-1)^2}$$

$$= \sqrt{0.0625}$$

$$= 0.0625$$

O2 and O5 :-

$$D = \sqrt{0^2 + 1^2 + 0^2 + 0^2 + (0-0.0625)^2}$$

$$= \sqrt{1.00390625}$$

$$= 1.00195$$

The elements within this have small distance, hence this will be one cluster.

Using same notation generate distance matrix:-

Step 4: 0	01	02	03	04	05
00					
01	0	1.697	0.063	1.616	1.663
02	1.697	0	1.732	1.423	0.063
03	0.063	1.732	0	1.647	1.697
04	1.616	1.423	1.647	0	1.417
05	1.663	0.063	1.697	1.417	0

Step 5: HAC

We use single linkage, where:-

$$D(A, B) = \min d(x, y)$$

btw members of two clusters.

merge 1

smallest distance:-

$$01 - 03 \Rightarrow 0.0625$$

By seeing smallest distance

$$C_1 = 01 + 03$$

$$C_2 = 02 + 05$$

$$C_3 = 04$$

$$C_1 = \{01, 03\} = 0.0625$$

$$C_2 = \{02, 05\} = 0.0625$$

$$C_3 = \{04\} = 0.0625$$

30-tee form one similar group

U2, U4

→ low usage

→ low login frequency

→ High purchase amount.

Step 3: compare two groups

features	U1 U3 U5 (features)	U2 U4
usage	high	low
login	4	4
purchase.	low	high

There is a clear difference b/w two groups.

Step 4: Define No. of clusters

since there are two distinct behavioural patterns
we choose $k=2$.

Therefore:

No. of clusters = 2

Steps: name the clusters

cluster 1 - highly Engaged

cluster 2 - less Engaged

Step 6: final result

$k=2$

$C1 = \{U1, U3, U5\}$

$C2 = \{U2, U4\}$

C104

min(03104)

min(0104, 103104)

min(16161, 1649)

C204

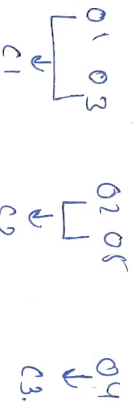
(02, 05)04

min((02104), 101041)

min(14321, 14119)

1616.

14119



Q3) Dataset: mobileApp User Behaviour

<u>User</u>	<u>Time</u>	<u>Log frequency</u>	<u>inApp purchase</u>
U1	120	18	2000
U2	60	5	5000
U3	140	22	1000
U4	90	4	6000
U5	130	25	1800

Step1 identify the features.

We have 3 features for each user:

- usage time (minutes)
- login frequency
- purchase amount (€).

Step2 Observe the Data.

look at the user's behavior

U1, U3, U5.

Pair	Distances
F1-F4	20.64
F2-F5	20.64
F1-F3	100.14
F2-F3	100.52
F3-F4	120.45
F3-F5	120.97
F4-F5	200.60
	241.37

HAC before normalization

using single linkage for smallest distance:-

$$F1-F4 = 20.64$$

$$\text{and } F2-F5 = 20.64$$

Therefore,

$$C1 = \{F1, F4\}$$

$$C2 = \{F2, F5\}$$

$$C3 = \{F3\}$$

Next, $\{F\}$ join

$C1$, because

$$\min[d(F3, F1), d(F3, F4)] = \min(100.14, 120.45) \\ = 100.14$$

so:

$$C1 = \{F1, F3, F4\}$$

finally, $\{F\}$ connects to $C2$ at 100.52, producing one cluster.

4) Dataset : Agriculture crop Monitoring

<u>Farm</u>	Rainfall (mm)	Soil fertility	Crop yield
F1	300	8	85
F2	500	4	76
F3	400	6	80
F4	280	9	90
F5	520	3	65

In Agriculture research center performs clustering on farm data before and after applying normalization. Interpret the difference observed in the cluster formation and improves the reliability of results. Explain how and improve the reliability of agriculture

Step 1:

Calculate distance before normalization.

We use Euclidean distance

$$d(P, Q) = \sqrt{\sum (P_i - Q_i)^2}$$

for example:

$$d(F1, F4) = \sqrt{(300-280)^2 + (8-9)^2 + (85-90)^2}$$

$$\sqrt{426} \approx 20.64$$

Similarity the important distance are.

Step 2:-

min-max normalization

The min-max formula

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

→ Rainfall, $x_{min} = 280$, $x_{max} = 320$

→ Soil fertility, $x_{min} = 3$, $x_{max} = 9$

→ Crop yield: $x_{min} = 65$, $x_{max} = 90$

<u>farm</u>	<u>Rainfall</u>	<u>Soil Fertility</u>	<u>Crop yield</u>
F1	0.0833	0.833	0.9000
F2	0.9167	0.1667	0.2000
F3	0.5000	0.8000	0.6000
F4	0.0000	1.0000	1.0000
F5	1.0000	0.000	0.000

Step 3:- Euclidean Distance

<u>pair</u>	<u>Distance</u>
F1-F4	0.2920
F2-F5	0.2934
F1-F3	0.5698
F2-F3	0.6669
F3-F4	0.8124
F3-F5	0.9294

Step 2:-

min-max normalization

The min-max formula

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

- Rainfall ; $x_{\min} = 280$, $x_{\max} = 520$
- soil fertility ; $x_{\min} = 3$, $x_{\max} = 9$
- Crop Yield : $x_{\min} = 65$, $x_{\max} = 90$

<u>Farm</u>	<u>Rainfall</u>	<u>Soil Fertility</u>	<u>Cropfield</u>
F1	0.0833	0.833	0.9000
F2	0.9167	0.1667	0.2000
F3	0.5000	0.5000	0.6000
F4	0.000	1.0000	1.0000
F5	1.0000	0.000	0.000

Step 3:- Euclidean Distance

<u>Pair</u>	<u>Distance</u>
F1-F4	0.2736
F2-F5	0.2734
F1-F3	0.5698
F2-F3	0.6669
F3-F4	0.8129
F3-F5	0.9299

Step 4: HAC clustering

At 3 clusters

$C1 = \{F1, F4\}$

$C2 = \{F2, F5\}$

$C3 = \{F3\}$

∴ The similar forms
are grouped
together.

Step 5: Interpretation

Before normalization rainfall dominates it because
its numerical range is much larger. After
normalization all features are scaled to 0-1,
giving rainfall, soil fertility and crop yield
balanced influence.

The final 3 clusters.

$\{F1, F4\}, \{F2, F5\}, \{F3\}$

5/a

Dataset Employee

Performance

Evaluation:

Employee productivity

Attendance

(%)

Skill
assessment
score

E1

85

92

84

E2

62

75

68

E3

78

88

72

E4

88

65

58

E5

93

96

90

im:
77

separation b/w different clusters. It can efficiently group employees based on the productivity attendance, and still assured.

Conclusion:

Best no. of clusters = 3.

The selected $c=3$ model can help the company identify different employee performance groups, such as high, medium, and low performers. This supports better performance evaluation, training, recognition, and employee development.